

Instructions

- Answer all eight questions. Show sufficient working for method marks.
- A calculator may be used unless a question states otherwise.
- Write clearly in the spaces provided. Additional pages may be attached.

1

Write pseudocode for a linear search and state its worst-case time complexity.

[5 marks]

$O(n)$

2

Explain the difference between a stack and a queue, giving one appropriate use for each.

[5 marks]

3

Trace the algorithm on the input [5, 2, 9, 1] and state the array after each pass.

[6 marks]

4

Convert the binary number to denary and explain the place-value method used.

[4 marks]

 $1011\ 0110_2$

5

Describe how a hash table resolves collisions, naming one strategy and its drawback.

[6 marks]

6

Write a function that returns the maximum value in a list without using a built-in max.

[5 marks]

7

Explain why binary search requires a sorted list, and state its complexity.

[5 marks]

$$O(\log n)$$

8

Compare compiled and interpreted execution, giving one advantage of each.

[5 marks]
